

Homework (Jalapeno)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 6)$
- (C) $x(x + 5) + 6$
- (D) $(x + 3)(x + 2)$
- (E) $(x + 1)(x + 4)$

Q2. What is the factored form of $2x^2 + 7x + 3$?

- (A) $(2x + 1)(x + 3)$
- (B) $(2x + 3)(x + 1)$
- (C) $2(x + 1)(x + 3)$
- (D) $(x + 1)(2x + 3)$
- (E) $2(x + 3)(x + 1)$

Q3. The ancient Egyptians used trinomials to calculate areas of triangles. Factor the trinomial $x^2 - 4x - 5$ to find the dimensions of a triangle.

- (A) $x(x - 4) - 5$
- (B) $(x - 5)(x + 1)$
- (C) $x(x + 1) - 4x - 5$
- (D) $(x + 1)(x - 5)$
- (E) $(x - 4)(x + 1)$

Q4. The ancient Babylonians used factoring trinomials to solve problems related to the construction of the Hanging Gardens. Factor $3x^2 + 2x - 1$.

- (A) $3(x^2 +$

$\frac{2}{3}x - 1)$

- (A) $(3x + 1)(x + 1)$
- (B) $3(x + 1)(x - \frac{1}{3})$
- (B) $(3x - 1)(x + 1)$
- (C) $3(x - \frac{1}{3})(x + 1)$

Q5. The ancient Greeks used algebraic methods to calculate the areas of their famous amphitheaters. Factor $x^2 + 2x - 6$.

- (A) $x(x + 2) - 6$
- (B) $(x + 3)(x - 2)$
- (C) $x(x - 2) + 2x - 6$
- (D) $(x + 2)(x - 3)$
- (E) $(x + 3)(x - 2)$

Q6. Factor the trinomial $x^2 - 3x - 4$.

- (A) $x(x - 3) - 4$
- (B) $(x - 4)(x + 1)$
- (C) $x(x + 1) - 3x - 4$
- (D) $(x + 1)(x - 4)$
- (E) $(x - 4)(x + 1)$

Q7. Factor $4x^2 + 12x + 5$.

- (A) $4(x^2 + 3x) + 5$
- (B) $4(x^2 + 3x + \frac{5}{4})$

- (B) $(2x + 1)(2x + 5)$
- (C) $2(2x^2 + 6x) + 5$
- (D) $(2x + 5)(2x + 1)$

Q8. The Rosetta Stone has an inscription that describes the construction of a temple. Factor the trinomial $2x^2 + x - 3$ to find the area of the temple.

- (A) $2(x^2 +$

$\frac{1}{2}x) - 3$

- (A) $(2x - 3)(x + 1)$
- (B) $2(x^2 + \frac{1}{2}x - \frac{3}{2})$
- (B) $(x + 1)(2x - 3)$
- (C) $(2x + 3)(x - 1)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $x^2 + 10x + 24$ and explain how the ancient Chinese used trinomials to calculate the dimensions of the Great Wall.

Q10. The ancient Romans used factoring trinomials to design their famous aqueducts. Factor $3x^2 - 2x - 5$ and describe how this factored form can be used to calculate the height of an aqueduct.

Chef's Tips

- Check for the correct factored form of the trinomial.
- Ensure that students provide a clear explanation of how the factored form can be used in real-world contexts.
- Emphasize the importance of factoring trinomials in various historical and ancient contexts.